

Outnumbered by the Invisible: Calculating the amount of Galactic Dark Matter



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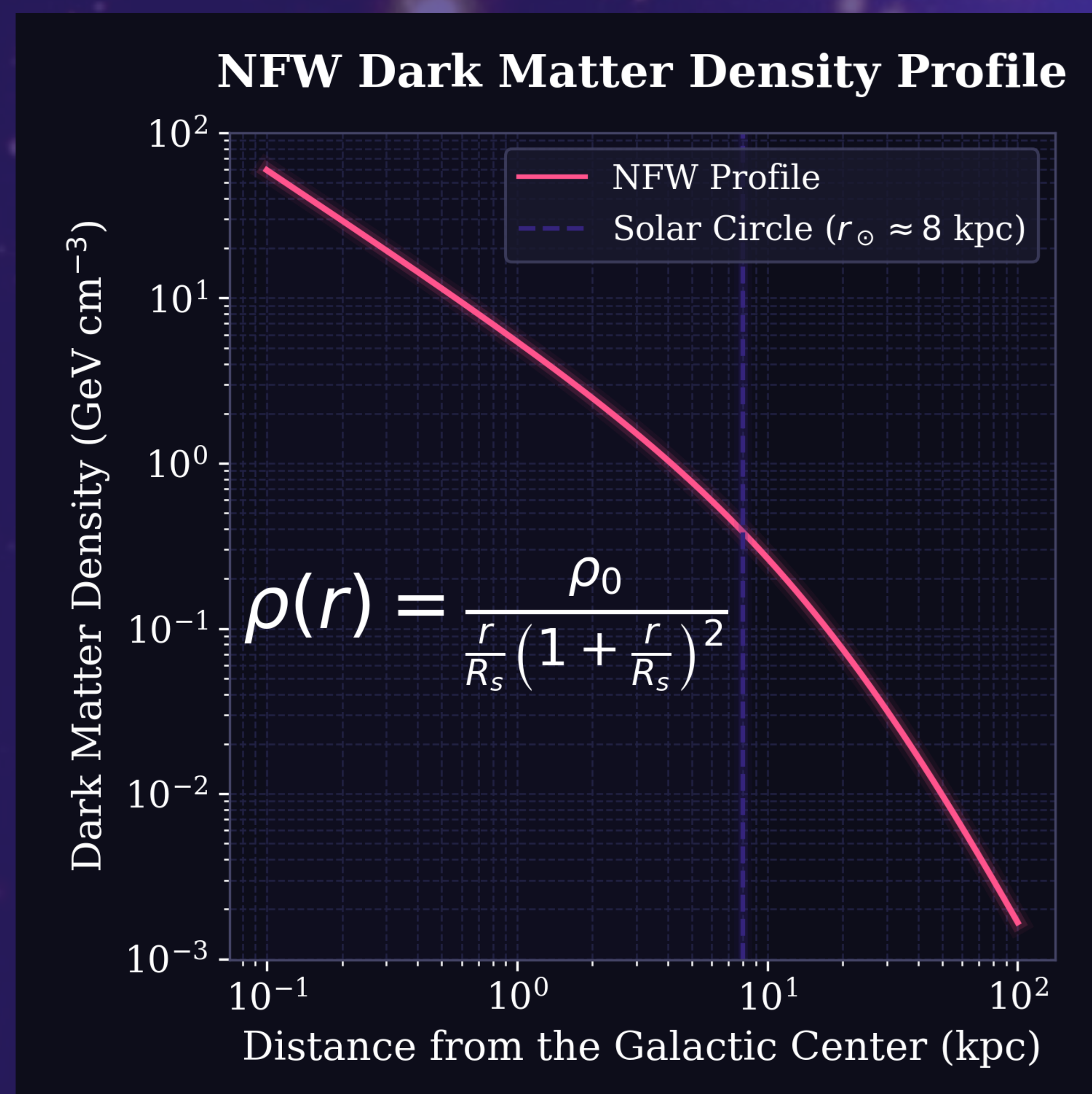
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Introduction

Dark matter composes ~85% of the matter in the universe [1], yet it cannot be directly observed. In this project, we utilize dark matter density and cored-NFW profiles to derive formulas and determine the mass of dark matter in the Galaxy and plot its distribution on a skymap.

NFW Profile

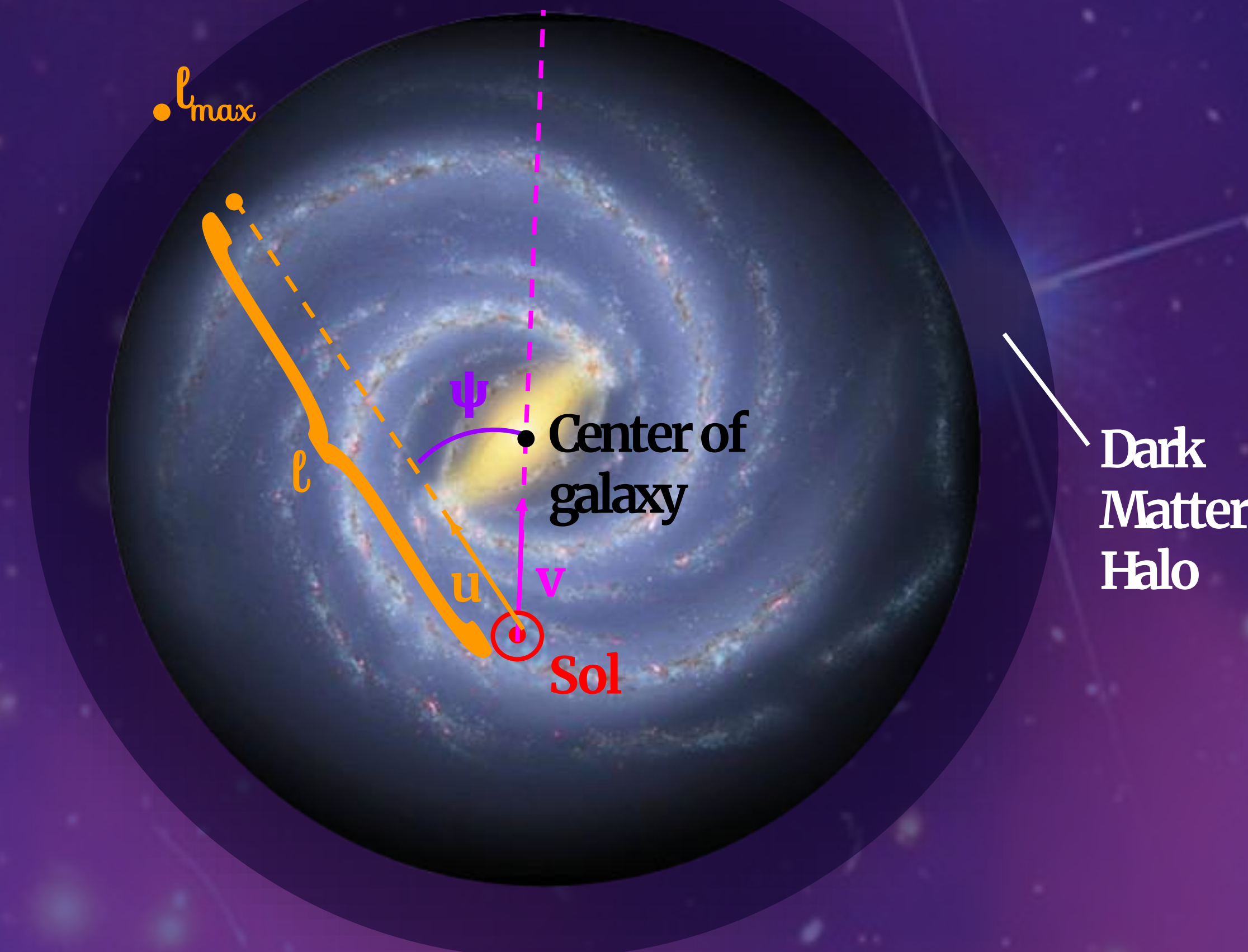
We considered a Navarro-Frenk-White [2] density profile, plotted below:



By numerically integrating this profile over the Galactic Halo, we obtain a dark matter mass of $M_{\text{dm}} = 1.8 \times 10^{12} M_{\odot}$.

Celestial Geometry

We considered calculating the dark matter column density along a line of sight through the galaxy.

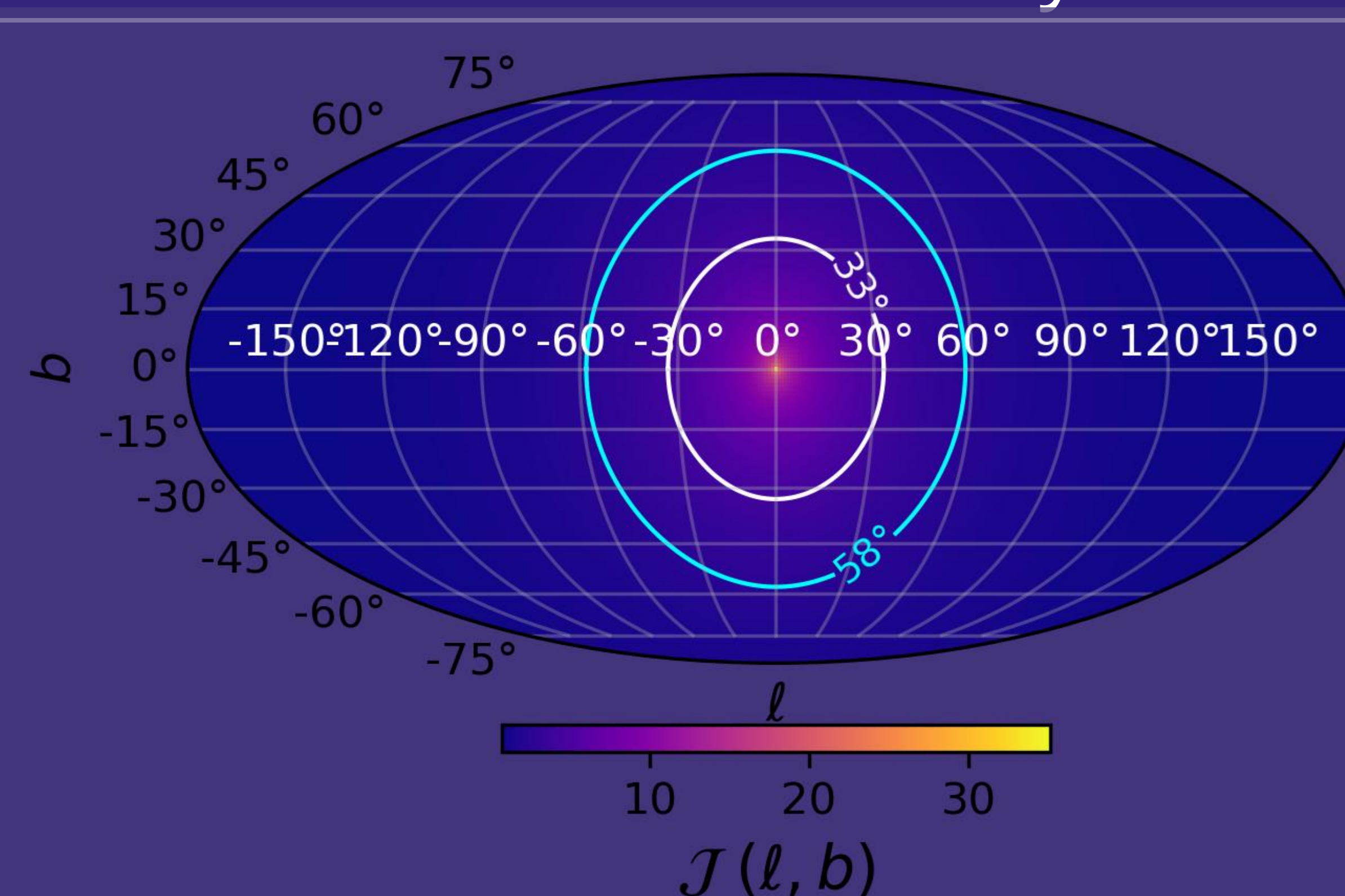


Using celestial geometry, we transform between coordinate systems to derive quantities used to calculate the dark matter column density [3]:

$$\mathcal{J}(\psi) = \frac{1}{R_{\text{sc}} \rho_{\text{sc}}} \int_0^{l_{\text{max}}(\psi)} \rho_{\text{dm}}(r(\psi, l)) dl$$

Mapping the J Factor

By integrating the NFW profile along lines of site, we obtained a map of the dark matter column density:



Discussion

During this project, I became familiar with:

- Numerical integrate in Python
- Create and customize plots in Python
- Account for numerical errors in integration
- Computational skills and data visualization, as well as applying theoretical knowledge to real-world calculations

Applications:

- Continued study of dark matter, galaxy formation, and cosmology
- Modeling how galaxies form and evolve
- Column density plots can be used to visualize the hidden structure of the cosmos and are involved with larger sky surveys, 3D reconstruction, and improved simulations

Acknowledgements

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References: [1] Bautista, Manuel. "Doe Explains...Dark Matter." *Energy.Gov*. www.energy.gov/science/doe-explainsdark-matter. Accessed 14 Apr. 2026. [2] Navarro, J. F., et al. "Simulations of X-ray clusters." *Monthly Notices of the Royal Astronomical Society*, vol. 275, no. 3, 1 Aug. 1995, pp. 720–740. <https://doi.org/10.1093/mnras/275.3.720>. [3] Zaidel, M., Murase, K., Zhang, B. (2022) Constraining Metastable Very Heavy Dark Matter using the Extragalactic Gamma-ray Background. *Pennsylvania State University McNair Scholars Program Journal*.